

René Enjilian

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Education

Technische Universität Berlin

M.Sc. in Computer Science Final grade: **1.1** (1.0 = best)

Thesis: Workload-aware and Reconfigurable LLM Serving [thesis] [code]

Berlin, Germany

Apr 2023 – June 2026

Technische Universität Berlin

B.Sc. in Computer Science Final grade: **2.2**

Thesis: Towards Single-Source Query Processing on Heterogeneous Hardware

Academic interruption: Oct 2019 – Mar 2021 due to full-time family caregiving responsibilities

Berlin, Germany

Oct 2016 – Oct 2022

Experience

Research Assistant

Sep 2023 – Aug 2025

DAMS Lab, BIFOLD, Technische Universität Berlin

Berlin, Germany

- Contributed to [Apache SystemDS](#), an open-source ML system for scalable data science.
- Implemented algebraic rewrites over the SystemDS compiler DAG to reduce compute and memory overhead.
- Developed runtime components for compressed and sparse matrix operations.
- Modernized the SystemDS GPU backend and developed custom JNI bindings for CUDA, cuSPARSE, and cuBLAS.
- Contributed ~30k LOC in Java, C++, and CUDA across compiler, runtime, and GPU backend components.
- Recognized by the Apache Software Foundation as an Apache SystemDS committer for sustained contributions across multiple system layers ([ASF ID: renjilian](#)).

Projects

Kairos – LLM Serving System | Python, vLLM, huggingface, FastAPI, ZeroMQ, Docker, CUDA [summary]

- Built Kairos, a workload-aware and reconfigurable LLM serving system on top of vLLM. [code]
- Kairos monitors requests and adapts model selection and placement to satisfy latency and accuracy objectives.
- Evaluates base, quantized (FP8/INT8/INT4/NF4), and smaller LLMs for the best latency/accuracy trade-off.
- Reconfigures model placement across GPU, CPU memory, and disk when a better configuration is found.
- Forked [vLLM](#) and extended its internals for explicit model weight movement across GPU, CPU, and disk. [code]
- These extensions add prefetching, eviction, and sleep/wake transitions for fast model swapping.

Apache SystemDS GPU Backend Modernization | Java, JCuda, CUDA, cuBLAS, cuSPARSE, cuDNN

- Migrated the SystemDS GPU backend to the CUDA 12.6 software stack (cuBLAS 12, cuSPARSE 12, cuDNN 9).
- Replaced deprecated APIs and integrated the required descriptors and pipelines following NVIDIA's porting guides.
- Achieved up to **3×** faster inference workloads. (**~3k LOC**) [code]

Native Java-CUDA Bindings for Apache SystemDS | Java, JNI, CUDA, C++, cuBLAS, cuSPARSE

- Developed native JNI bindings for calling CUDA, cuSPARSE, and cuBLAS from the Java-based SystemDS runtime.
- Reduced reliance on JCuda by exposing custom Java interfaces to NVIDIA GPU libraries. (**~11k LOC**) [code]

Apache SystemDS Matrix Compression Runtime | Java

- Implemented the Compressed Sparse Column (CSC) format for sparse matrix operations. (**~2k LOC**) [code]
- Implemented the Modified CSC (MCSC) format for sparse matrix operations. (**~1.2k LOC**) [code]

SLAB Integration for Apache SystemDS | Java, Python, Apache Spark

- Reimplemented the [Scalable Linear Algebra Benchmark](#) (SLAB) in Apache SystemDS. (**~2.3k LOC**) [code]

I-JEPA Reproduction and Analysis | Python, PyTorch, SLURM

- Pretrained [I-JEPA](#) from scratch on ImageNet-1K using 8 NVIDIA A100 GPUs on a ViT-Base backbone. [code]
- Systematically evaluated I-JEPA's embedding-invariance claim via cosine similarity under various data augmentations.
- Investigated I-JEPA internals with occlusion analysis, neuron activations, and concept activation vectors. [report]

Technical Skills

Languages: Python, CUDA, C/C++, Java, SQL, JavaScript, R

Frameworks / Libraries: vLLM, PyTorch, huggingface, NumPy, JNI, pandas, FastAPI, ZeroMQ

GPU Computing: cuBLAS, cuSPARSE, cuDNN, JCuda, nvidia-smi, Nsight

Tools / Infrastructure: Git, Docker, Kubernetes, SLURM, Apptainer, Linux